

# Object Detection Extension

<https://github.com/aiyshwary/Object-Detection-Extension-in-Rapid-Miner>

[Python](#)

[PyTorch](#)

[ONNX Runtime](#)

[RapidMiner](#)

[Object Detection](#)

## OVERVIEW

A RapidMiner Studio extension that brings transfer learning-based object detection into a no-code environment, supporting fine-tuning and inference with Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet models.

## IMPACT

Developed two RapidMiner operators for object detection — `FineTuneObjectDetectionModel` and `ObjectDetectionModelInference` — enabling users to train and deploy detection models within a visual workflow without writing code.

## TECHNICAL WALKTHROUGH

### Object Detection Extension

#### Problem Statement

Different object detection use cases require different trade-offs between accuracy, speed, and computational cost. A single model cannot satisfy all scenarios.

#### Approach

Developed a unified object detection extension using PyTorch, supporting multiple architectures:

**Faster R-CNN**

**FCOS**

**SSD**

**SSD Lite**

**RetinaNet**

YOLO (extended work)

#### Training

**Dataset annotated using XML bounding boxes**

**Transfer learning using COCO pre-trained weights**

**Fully configurable training pipeline:**

**Model selection**

**Batch size**

**Epochs**

**Learning rate**

**Momentum**

**Weight decay**

**Evaluation metrics:**

IoU (Intersection over Union)

## Precision

Recall (per class)

## Inference

Users can choose any trained or pre-trained model

## Output includes:

Image with bounding boxes, labels, and confidence scores

## Tabular output with detected objects

Results also generated in base64 format for easy deployment and integration

## Key Observations

### Faster R-CNN performed best for accuracy

SSD and SSD Lite were faster but had lower recall

### RetinaNet handled class imbalance well

### Model choice depends on application requirements

## KEY DETAILS

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- `FineTuneObjectDetectionModel` accepts image and annotation directories (YOLO/Pascal VOC) and outputs model weights and class list CSV.
- `ObjectDetectionModelInference` runs predictions on new images and returns bounding boxes and class labels as a `DataFrame`.
- Supports five architectures: Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet for varying speed-accuracy trade-offs.
- Configurable for CPU-only and GPU (CUDA cu118) environments; installed as a JAR extension in RapidMiner Studio.
- Enables domain-specific object detection with minimal ML expertise using a drag-and-drop process designer.